

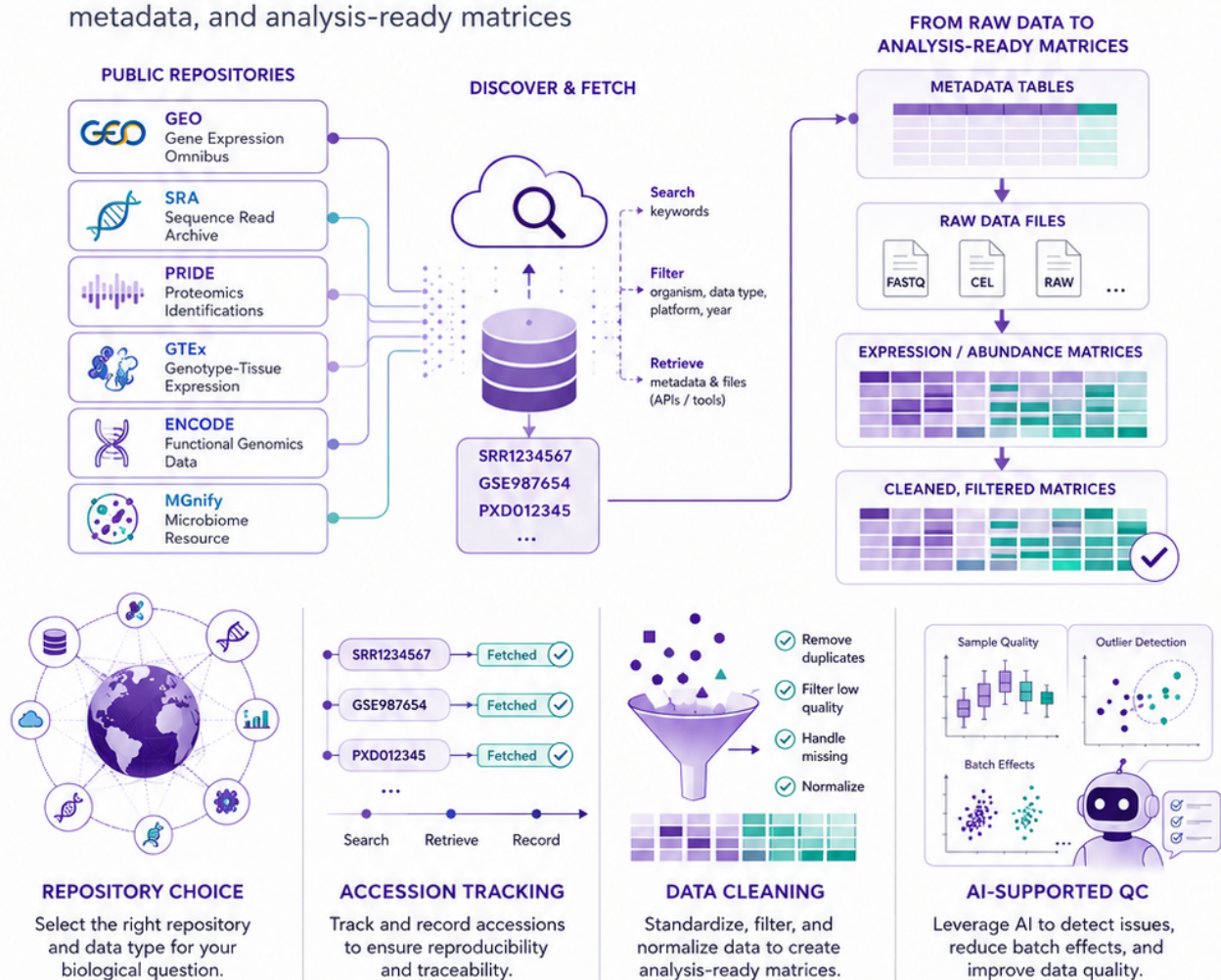


BIOINFORMATICS: FROM MULTI-OMICS DATA TO DISCOVERY

WEEK 3 READING MATERIAL

Public Repositories, Data Fetching, and Cleaning

From biological questions to traceable public datasets, metadata, and analysis-ready matrices



Fall 2026

Two 40-Minute Undergraduate Sessions

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Week 3 learning architecture

From a biological question to a traceable, analysis-ready public dataset.

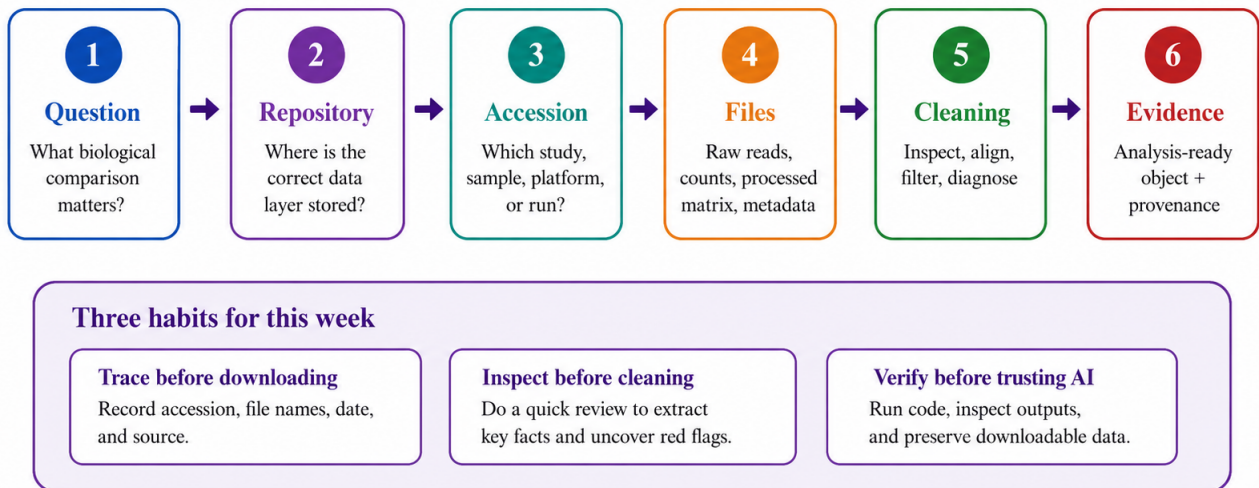


Figure 1. Week 3 learning architecture. Students move from a biological question to a traceable, inspected, and analysis-ready public dataset.

How to use this handout: Read Pages 3–12 before Session 1 and Pages 13–27 before Session 2. The one-hour homework on Page 28 consolidates retrieval, cleaning, AI verification, and reproducibility. Page 29 lists likely quiz concepts for Week 4.

Learning objectives, deliverables, and class schedule

By the end of Week 3, students should be able to identify the correct public repository and accession type, judge whether a dataset is scientifically usable, retrieve GEO records in R, align expression data with metadata, diagnose common quality problems, select an appropriate normalization strategy, and document AI-assisted work transparently.

Competency	What students should be able to do	Concrete deliverable
Repository navigation	Match GEO, SRA/ENA, GDC/TCGA, GTEx, ENCODE, PRIDE, MetaboLights, or MGnify to a data need.	Repository choice with justification
Accession literacy	Interpret GSE, GSM, GPL, GDS, BioProject, SRP, SRX, and SRR identifiers.	Accession crosswalk
Dataset triage	Evaluate biological fit, replication, metadata, files, provenance, access, and confounding.	12-point triage scorecard
Programmatic retrieval	Use GEOquery to obtain a GEO study object and inspect supplementary files.	R script + manifest
Data cleaning	Check structure, alignment, missingness, duplicates, low-information features, outliers, and batch.	Cleaned object + QC figure
AI/agent practice	Use plan-first prompting and verify accessions, file meaning, code execution, and outputs.	Prompt + verification log
Reproducibility	Preserve raw data and record versions, decisions, dates, and paths.	Project folder + sessionInfo()

Two-class schedule

Session	Focus	In-class activity	Connection to homework
Session 1: 40 min	From research question to the correct repository, accession, and data layer.	GEO page triage and accession-mapping challenge.	Choose and justify one candidate dataset.
Session 2: 40 min	Programmatic retrieval, metadata alignment, cleaning, QC, normalization, AI/agent verification, and reproducibility.	Live GEOquery + matrix/metadata debugging clinic.	Submit one script, one QC figure, and a decision log.

Core rule: Public data are not automatically analysis-ready. Repository records, files, metadata, and software outputs must be interpreted before they can support a scientific claim.

Session 1 | Public data are reusable evidence, not anonymous files

The scientific value of a public dataset depends on provenance, design, metadata, and the match to your question.

1.1 Open data changes what an undergraduate can investigate

A public repository allows a student to revisit an experiment without repeating sample collection. The opportunity is substantial: on 21 July 2026, the GEO home page listed 289,760 Series records and 8,622,867 Samples [1]. The important skill is therefore not simply “finding data,” but selecting a study that can answer a defined question and preserving enough provenance to reproduce the retrieval.

Reuse can take several forms: reproducing a published analysis, testing a new contrast in the same study, combining compatible cohorts, benchmarking a method, or generating a hypothesis for later experimental validation. None of these uses remove the need to understand how the samples were generated.

1.2 FAIR principles as a practical checklist

FAIR element	Student interpretation	Week 3 action
Findable	Persistent identifiers and searchable metadata make a dataset discoverable.	Record GSE/SRP/BioProject and DOI.
Accessible	Access conditions and retrieval methods are explicit.	Note open versus controlled access.
Interoperable	Standard formats and identifiers allow software and datasets to connect.	Identify genome build, gene IDs, file type, and sample IDs.
Reusable	Metadata, methods, licensing, and provenance support a new analysis.	Document study design, files, filters, and limitations.

Interpretation boundary: A public dataset can provide evidence about associations and reproducibility. It does not automatically prove causality, generalizability, or the correctness of the original analysis.

1.3 Provenance questions to ask before downloading

- Who generated the data, for what biological question, and under which protocol?
- Which organism, tissue, disease state, treatment, time point, or cell type is represented?
- What level of data is public: raw reads, feature counts, normalized values, or derived result tables?
- Which reference genome, annotation, platform, and software pipeline were used?
- Can each sample be linked to complete group labels, covariates, and batch variables?

The public repository ecosystem

Public repositories form a connected data ecosystem

Choose the repository according to the biological object and the downstream analysis.

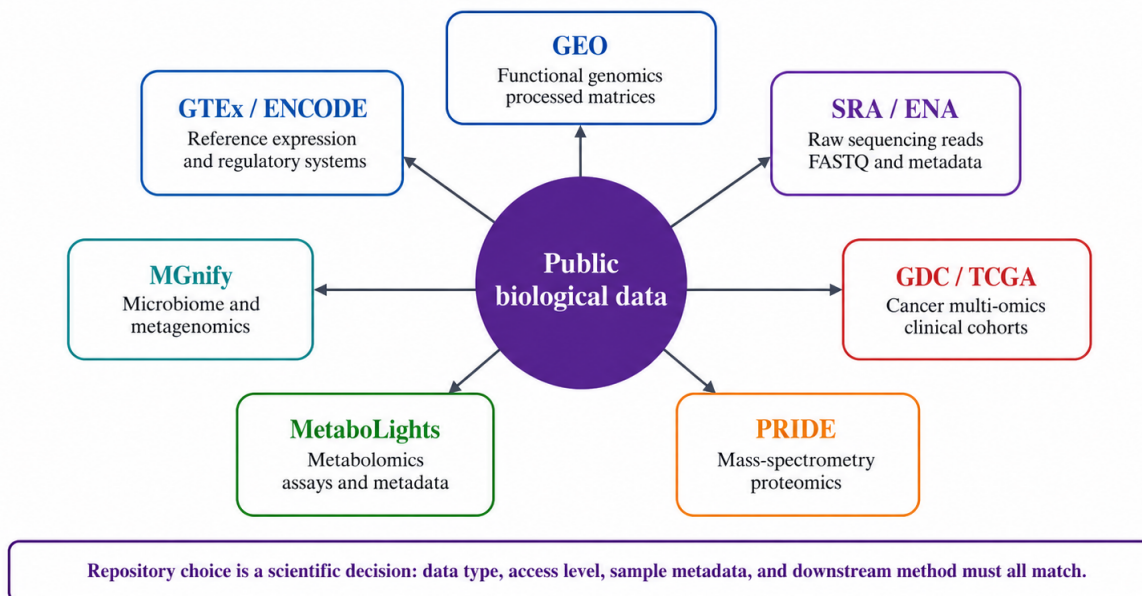


Figure 2. Public repositories are complementary. The correct choice depends on assay type, processing level, access conditions, and the downstream question.

Repository	Typical content	Best use in this course	Important caution
GEO	Functional genomics studies, processed matrices, sample metadata, and links to raw reads.	Expression, epigenomic, and single-cell study discovery.	A Series may contain multiple platforms or incomparable assay types.
SRA / ENA	Raw sequencing reads and run metadata.	Reprocessing FASTQ from the beginning.	Computationally intensive; sample-to-run mapping must be checked.
GDC / TCGA	Cancer genomic and clinical data.	Large multi-omics cancer cohorts.	Some files are controlled access; identifiers and versions matter.
GTEx / ENCODE	Reference tissue expression and regulatory genomics.	Normal-tissue comparison and regulatory context.	Study design differs from disease cohorts.
PRIDE	Mass-spectrometry proteomics files and metadata.	Proteomics reanalysis.	Raw vendor files and search parameters may require specialist tools.
MetaboLights	Metabolomics studies, assays, and metadata.	Metabolite data reuse.	Compound identification confidence varies.

Repository	Typical content	Best use in this course	Important caution
MGnify	Microbiome and metagenomics analyses and study records.	Community and functional microbiome datasets.	Processing pipelines may differ from the original publication.
NMDC	National Microbiology Data Center	Host of Chinese investigators' microbiology-related data	Inquiry of the meta data for samples

Repository selection rule: Choose the repository that stores the biological object and data level required by your planned method—not the repository with the easiest download button.

Choosing a repository by question and assay

1.4 Start from the biological object

Question or data need	Primary repository	Possible companion resource	Likely data object
Bulk or single-cell transcriptomics	GEO	SRA/ENA, Single Cell Portal	Series matrix, count matrix, FASTQ, H5/MTX
Cancer RNA-seq + mutations + clinical outcomes	GDC/TCGA	GEO, cBioPortal	Counts, MAF, clinical tables
Healthy-tissue expression reference	GTEx	GEO	TPM/counts, tissue metadata
TF binding or chromatin accessibility	ENCODE / GEO	SRA	BED peaks, bigWig, FASTQ
Shotgun metagenomics	MGnify / SRA / ENA	GEO for paired host data	FASTQ, taxonomic/functional profiles
Mass-spectrometry proteomics	PRIDE	UniProt	Raw spectra, mzML, result tables
Metabolomics	MetaboLights	HMDB / MetaboAnalyst resources	mzML, feature tables, metadata
Human genotype-phenotype data	dbGaP / EGA	GEO for public summaries	Controlled-access genotype and phenotype files

1.5 Open and controlled access are different scientific workflows

Open-access repositories allow direct retrieval, but human genomic and clinical data may require institutional approval, a data-use agreement, and secure computing. “Publicly described” does not always mean “freely downloadable.” Students should never try to bypass access controls or move controlled data into unapproved tools.

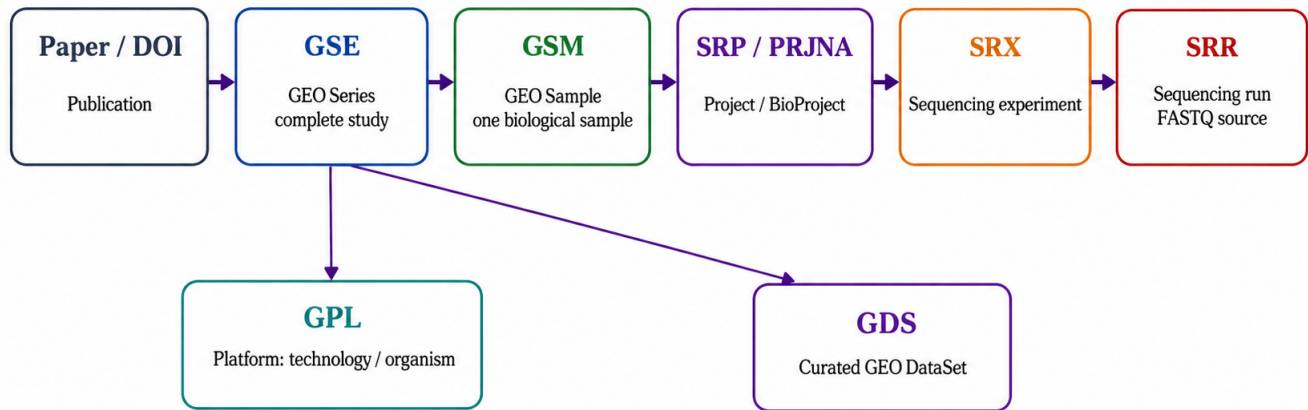
Data situation	Allowed student action	Not allowed
Open GEO/SRA study	Download public metadata and files; cite accession and publication.	Assume the data are free of privacy or consent limitations.
Controlled human data	Read public study description; discuss an analysis plan; use approved access if provided by the institution.	Upload protected data to a public AI system or personal cloud storage.
Unclear license or consent	Pause and ask the instructor or data custodian.	Proceed because the file can technically be downloaded.

Ethics checkpoint: Data governance is part of bioinformatics competence. Technical access does not replace permission, consent, or data-use restrictions.

GEO accession grammar

Accession prefixes describe different objects in the data chain

A paper may link one study to samples, sequencing experiments, runs, platforms, and files.



Practical rule: start from GSE for the study, use GSM for sample metadata, and follow SRA/BioProject links when raw reads are required.

Figure 3. Accession prefixes identify distinct objects. A single study can connect a publication, GEO Series, samples, platform, BioProject, sequencing experiments, and runs.

Object	Prefix	Meaning	Typical student use
Series	GSE	A complete study, sometimes spanning more than one platform.	Primary entry point for a study.
Sample	GSM	One submitted biological sample or library record.	Read sample-specific metadata and file links.
Platform	GPL	Technology and organism-specific platform definition.	Map probes/features and understand measurement technology.
DataSet	GDS	NCBI-curated dataset assembled from GEO records.	Curated exploration; less common for current workflows.
BioProject / SRA project	PRJNA / SRP	Project grouping raw sequence experiments.	Connect a GEO study to raw reads.
Experiment	SRX	Sequencing experiment or library object.	Understand library strategy and run grouping.
Run	SRR	One sequencing run, usually downloadable as FASTQ.	Raw-read retrieval and checksums.

Practical rule: Use GSE to understand the study, GSM to understand samples, GPL to understand the measurement platform, and SRA/BioProject links when raw sequence reprocessing is required.

From a paper to the correct public files

1.6 A reproducible retrieval chain

1. Locate the paper's Data Availability statement and record the DOI and accession.
2. Open the GSE page and confirm title, organism, experimental design, platform, and sample count.
3. Read the GSM table and determine whether the intended groups are genuinely comparable.
4. Inspect "Supplementary file," "Series Matrix File," and SRA/BioProject links.
5. Record exact file names, sizes, download date, and whether the values are raw counts, normalized values, or derived results.
6. Read the methods and platform annotation before assigning gene identifiers or groups.

1.7 A Series page contains several kinds of evidence

Page element	What it tells you	Question to ask
Summary / Overall design	Biological purpose, groups, interventions, and sample structure.	Does the design match my research question?
Samples (GSM)	Sample-level source, characteristics, library strategy, and files.	Can every matrix column be mapped to one biological sample?
Platform (GPL)	Technology, organism, feature IDs, and annotation.	Are rows genes, transcripts, probes, or peaks?
Series Matrix	Submitter-supplied processed matrix and metadata.	Are values already normalized or log-transformed?
Supplementary files	Counts, matrices, peak files, H5/MTX, or derived results.	Which file is closest to the required input for my method?
SRA / BioProject links	Raw reads and run-level metadata.	Do I need full reprocessing, and can I map runs to samples?
Publication link	Methods, context, and original conclusions.	What processing assumptions will I inherit?

Common error: Downloading the first TXT file without reading its description. A file can contain normalized expression, selected genes, or a differential-result table rather than the feature-by-sample matrix you need.

Data levels and file formats

Public studies provide multiple data layers with different analytical meanings

Processed data are faster to use; raw data provide greater control but require more computation and documentation.

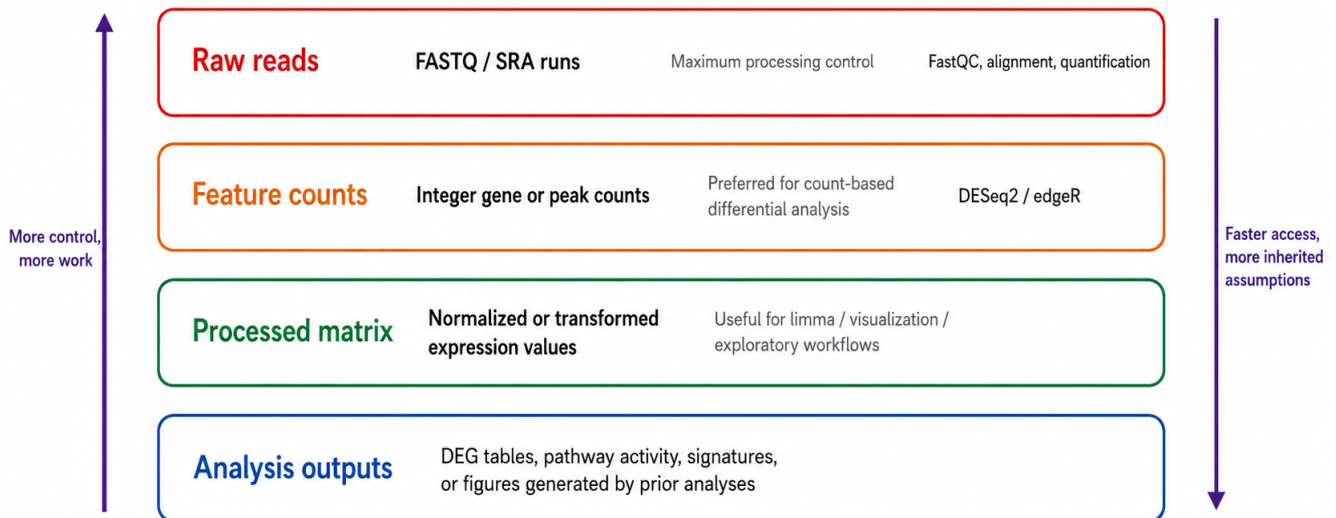


Figure 4. Data layers trade processing control for convenience. The appropriate level depends on the downstream question and method.

File or value type	What it usually contains	Suitable uses	Major risk
FASTQ	Raw sequencing reads with quality scores.	Full reprocessing, benchmarking, alternate references.	Large compute/storage burden and run-to-sample mapping.
BAM / SAM	Aligned reads.	Coverage, splice, or variant analyses when alignment is trusted.	Reference and aligner assumptions are inherited.
Raw integer counts	Feature-by-sample counts.	DESeq2/edgeR differential analysis and QC transformations.	Must be unnormalized integers with correct metadata.
TPM / FPKM	Length- and library-adjusted expression.	Within-sample ranking and some visual comparisons.	Not valid as raw counts for DESeq2.
Microarray expression	Usually normalized, often log2-scale probe intensities.	limma, PCA, clustering, signature analysis.	Probe annotation and preprocessing must be understood.

File or value type	What it usually contains	Suitable uses	Major risk
H5 / MTX / loom	Sparse matrices and metadata for single-cell assays.	SingleCellExperiment/Seurat import.	Cell and feature metadata may be stored separately.
DEG or pathway table	Prior analysis results.	Context, hypothesis generation, result comparison.	Cannot replace the underlying matrix for independent analysis.

Critical distinction: “Processed” does not mean “incorrect,” and “raw” does not mean “better.” The correct data level is the one that preserves the information required by the planned model.

Dataset triage before download

A dataset should pass biological, design, metadata, and file checks before download

A polished GEO page does not guarantee that the study can answer your question.

Biological fit	Organism, tissue, cohort, comparison	0 fails	1 unclear	2 passes
Replication	Has per-group sample size and enough groups?	0 fails	1 unclear	2 passes
Metadata	Groups, covariates, batch, identifiers	0 fails	1 unclear	2 passes
Files	Counts or usable processed matrix available	0 fails	1 unclear	2 passes
Provenance	Publication, accession, platform, genome build	0 fails	1 unclear	2 passes
Access & ethics	Open, controlled access, and permitted use	0 fails	1 unclear	2 passes
Recommended threshold: proceed only when no critical criterion fails and the total score is at least 9 of 12.				

Figure 5. A 12-point dataset triage score prevents students from committing to a study that cannot answer the intended question.

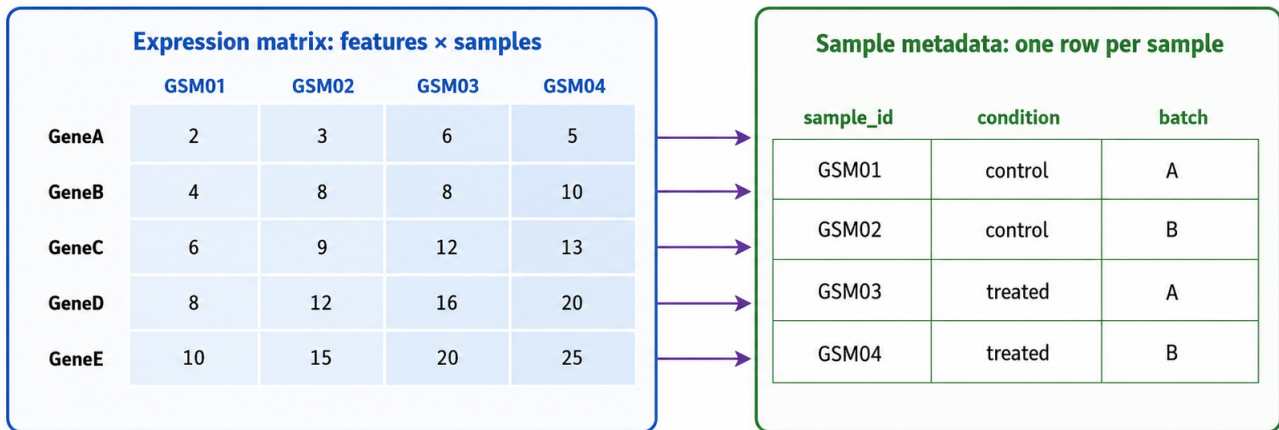
Criterion	2 points: passes	1 point: uncertain	0 points: fails
Biological fit	Organism, tissue, condition, and comparison match.	One element requires compromise.	Question and study design do not match.
Replication	Independent biological replicates and adequate groups.	Small or unbalanced groups.	No replication or pseudoreplication.
Metadata	Groups, covariates, batch, and sample IDs are usable.	Some variables need manual reconstruction.	Critical labels are missing or inconsistent.
Files	Required matrix/counts and metadata are public.	Only raw reads or awkward individual files.	Only derived results or unusable values.
Provenance	Publication, platform, genome build, and methods are clear.	One processing detail is unclear.	No traceable methods or annotation.
Access and ethics	Use is permitted and compatible with course tools.	Access conditions need clarification.	Use would violate restrictions or security.

Decision: Proceed when no critical criterion scores 0 and the total is at least 9/12. A lower score can still support a classroom demonstration, but its limitations must be explicit.

Metadata is the experimental design

Expression values and metadata must refer to the same samples in the same order

Many downstream errors are caused by silent misalignment rather than by incorrect statistics.



Required check:

`identical(colnames(counts), rownames(metadata))` must be TRUE before modeling.

Figure 6. Expression columns and metadata rows must refer to the same samples in the same order before any model is constructed.

1.8 Build a sample dictionary

Field	Example	Why it matters
sample_id	GSM2332515	Primary key linking matrix and metadata.
subject_id	Donor_01	Identifies repeated measures or paired samples.
condition	IRI_positive / IRI_negative	Defines the biological comparison.
time_point	pre / post reperfusion	Captures longitudinal or paired design.
batch	sequencing_run_A	Technical covariate or confounder.
tissue / cell type	liver biopsy	Defines biological source.
sex / age / stage / treatment	study-specific values	Potential covariates and inclusion criteria.
platform / genome build	HiSeq 2000 / GRCh38	Interprets feature identifiers and processing.

R checkpoint: align metadata to the matrix explicitly

```
stopifnot(!anyDuplicated(colnames(expr)))
stopifnot(!anyDuplicated(metadata$sample_id))
rownames(metadata) <- metadata$sample_id
metadata <- metadata[colnames(expr), , drop = FALSE]
stopifnot(identical(colnames(expr), rownames(metadata)))
```

Silent failure to avoid: If metadata are in the wrong order, R may still run and produce a valid-looking—but biologically incorrect—result.

Confounding, pairing, and pseudoreplication

1.9 The analysis unit is usually the biological replicate

Technical replicates, sequencing lanes, cells from the same patient, or repeated measurements are not automatically independent biological samples. Treating them as independent inflates sample size and can produce overconfident statistics. Before cleaning, identify the subject, sample, library, and run structure.

Design pattern	Correct interpretation	Potential model or action
Multiple sequencing runs for one library	Technical subdivisions of one sample.	Merge counts or model according to the quantified pipeline; do not count as separate subjects.
Pre/post biopsies from the same donor	Paired repeated measurements.	Include subject/blocking in design or use paired comparison.
Thousands of single cells from 6 patients	Cells are nested within patients.	Use sample-level/pseudobulk inference for group comparisons.
Batch perfectly matches condition	Batch and biology are confounded.	No statistical method can fully separate them; restrict claims or find another dataset.
Tumor and normal from different hospitals	Site may drive the apparent difference.	Include site if estimable; seek balanced or independent validation.

1.10 Dataset selection red flags

- The study has one biological sample per group but many technical files.
- Disease status is encoded inconsistently across title, characteristics, and publication.
- All controls were processed in one batch and all cases in another.
- The matrix contains sample names that cannot be matched to GSM records.
- Only a list of significant genes is provided.
- The platform annotation is outdated or gene identifiers cannot be traced.
- The publication's sample count does not match the repository files without explanation.

Scientific judgment: A dataset can be valuable despite imperfections, but the question and claims must be narrowed to what the design can actually support.

Session 1 activity | GEO dataset triage workshop

Work in pairs. Select one of the candidate studies supplied by the instructor, or use a GSE identified during Week 2. Spend 12 minutes on the repository page and 8 minutes defending your decision to another pair.

Step	Student action	Evidence to record
1	State the Week 2 data-driven question.	One sentence with population/system, comparison, outcome, and data type.
2	Confirm study identity.	GSE, title, organism, publication, and platform.
3	Map accessions.	At least one GSM and any linked BioProject/SRP/SRR.
4	Identify available data layers.	Series matrix, supplementary counts, raw reads, or single-cell files.
5	Reconstruct the design.	Groups, n, subject/paired structure, major covariates, batch.
6	Score the dataset.	0–2 for each of the six triage criteria.
7	Decide.	Proceed, proceed with limitations, or reject; justify in two sentences.

Mini-case: GSE87487

GSE87487 is a human bulk RNA-seq study of 20 liver allograft biopsies, with pre- and post-reperfusion sampling and a 3.7 MB supplementary count matrix. The Series links to SRA project SRP090633 [6]. It is useful for teaching because the repository exposes processed counts, raw-read links, and a paired biological design.

Question	Evidence from the record	Student decision
Are the values counts?	Supplementary file is named GSE87487_counts.20samples.txt.gz; methods used featureCounts.	Verify integer values after download.
Are samples independent?	Pre/post biopsies imply repeated observations.	Preserve subject pairing in metadata and design.
Can all 20 samples be compared directly?	IRI status and pre/post status create multiple factors.	Define a specific contrast before analysis.
Is full raw-read reprocessing possible?	Raw data link to SRA is present.	Possible, but beyond a one-hour homework.

Exit ticket: Write one sentence explaining why accession literacy and metadata inspection must occur before code is written.

Session 2 | Reproducible data fetching begins with a project structure

2.1 Separate immutable input from generated output

A reproducible project keeps the original download unchanged and writes all derived objects to new paths. This prevents accidental overwriting and makes it possible to audit every cleaning decision.

Folder	Contents	Rule
data_raw/	Original files exactly as downloaded.	Read-only; never edit or rename without preserving the source name.
metadata/	Sample manifest and hand-curated design table.	Every row has a stable sample_id.
R/	Numbered scripts: fetch, inspect, clean, QC.	Scripts run from the project root and avoid hidden manual steps.
data_clean/	Filtered matrices and serialized R objects.	File name states processing stage; never replace raw files.
results/	QC figures, logs, summaries, and later analysis outputs.	Everything can be regenerated by scripts.
README.md	Question, accession, file meaning, run order, and limitations.	Another student should be able to reproduce the workflow.
renv.lock / sessionInfo.txt	Software environment and package versions.	Generated after the workflow runs successfully.

Minimal Week 3 project tree

```

project/
├── README.md
├── data_raw/
├── metadata/
├── R/
│   ├── 01_fetch.R
│   ├── 02_inspect_clean.R
│   └── 03_qc.R
├── data_clean/
├── results/
├── renv.lock
└── sessionInfo.txt

```

Working-directory rule: Use an RStudio Project and relative paths such as `file.path("data_raw", filename)`. Avoid `setwd()` calls that only work on one computer.

Programmatic access with GEOquery

2.2 Install once, then retrieve by accession

Download and inspect a GEO Series

```
if (!requireNamespace("BiocManager", quietly = TRUE)) {
  install.packages("BiocManager")
}
BiocManager::install("GEOquery")

library(GEOquery)
library(SummarizedExperiment)

# A GSE may contain more than one platform.
gse <- getGEO("GSE39582", GSEMatrix = TRUE, getGPL = FALSE)
length(gse)
se <- gse[[1]]
class(se)
dim(se)
```

Current GEOquery documentation returns a list with one Bioconductor object per platform; current versions may return a SummarizedExperiment, while older teaching code often uses ExpressionSet [4,5]. Students should inspect class() rather than assume an object type.

Extract matrix, sample metadata, and feature annotation

```
# Current SummarizedExperiment accessors
expr <- assay(se)
metadata <- as.data.frame(colData(se))
features <- as.data.frame(rowData(se))

# Compatibility with a legacy ExpressionSet object
if (inherits(se, "ExpressionSet")) {
  expr <- Biobase::exprs(se)
  metadata <- Biobase::pData(se)
  features <- Biobase::fData(se)
}

dim(expr)
head(metadata[, intersect(c("title", "geo_accession", "source_name_ch1"),
  colnames(metadata)), drop = FALSE])
```

Verification: After getGEO(), confirm study title, platform, sample count, object class, matrix dimensions, and sample identifiers. A successful download is not proof that you selected the correct platform or value type.

Inspect supplementary files before downloading them

2.3 List the file inventory first

Inspect, then retrieve supplementary files

```
library(GEOquery)

supp <- getGEOSuppFiles("GSE87487", fetch_files = FALSE)
supp[, c("fname", "url", "size")]

# After confirming the intended file and its size:
dir.create(file.path("data_raw", "GSE87487"),
           recursive = TRUE, showWarnings = FALSE)
getGEOSuppFiles("GSE87487",
                baseDir = "data_raw",
                makeDirectory = TRUE)

list.files(file.path("data_raw", "GSE87487"), full.names = TRUE)
```

2.4 Create a download manifest

Manifest field	Example	Why it matters
accession	GSE87487	Persistent study identifier.
file_name	GSE87487_counts.20samples.txt.gz	Preserves the repository meaning.
source_url	Recorded from getGEOSuppFiles().	Allows retrieval to be repeated.
download_date	2026-07-21	Public files can be updated or replaced.
file_size	3.7 MB	Detects incomplete or unexpected downloads.
checksum	MD5 or SHA-256	Confirms file integrity and identity.
value_type	raw integer gene counts	Determines valid downstream methods.
genome / annotation	hg38; featureCounts	Interprets feature identifiers.
notes	20 samples; paired pre/post design	Captures scientific constraints.

Optional integrity check

```
tools::md5sum(list.files(file.path("data_raw", "GSE87487"),
                           full.names = TRUE))
```

Download discipline: Large raw-read files should not be downloaded during class unless the exercise requires them. Begin with metadata and file inventory; then retrieve only the data layer needed for the question.

Microarray and RNA-seq require different cleaning logic

2.5 Do not mix platform-specific workflows

Property	Microarray processed matrix	RNA-seq raw count matrix
Measurement	Probe intensity after array preprocessing.	Integer reads/fragments assigned to features.
Typical GEO source	Series Matrix / submitter normalized values.	Supplementary count matrix or NCBI-computed counts.
Common value range	Often log2-scale continuous values.	Non-negative integers with many zeros.
Feature identity	Probe IDs that may map many-to-one to genes.	Gene/transcript IDs, often Ensembl with versions.
Low-information filtering	Low variance or non-expressed probes, depending on purpose.	Low counts using count-aware threshold.
Differential model	limma.	DESeq2 or edgeR using raw counts.
Visualization transform	Values may already be log-normalized.	VST/rlog/log-CPM after count import.
Typical mistake	Applying DESeq2 to normalized intensities.	Providing TPM/FPKM as raw counts.

Worked teaching examples

Accession	Assay	Why used in Week 3	Important boundary
GSE39582	Affymetrix microarray; colorectal cancer cohort.	Demonstrates Series Matrix retrieval, platform annotation, phenotype data, and limma-oriented processing.	It is not RNA-seq; count filters and DESeq2 are inappropriate.
GSE87487	Bulk RNA-seq; 20 human liver allograft biopsies.	Demonstrates supplementary integer counts, SRA linkage, and paired metadata.	The pre/post and IRI factors require an explicit design; do not treat all columns as independent.

Instructor correction to the earlier handout: GSE39582 is a microarray study. The redesigned material uses it only for the processed microarray branch and uses a separate RNA-seq count example to teach count-specific cleaning.

Structural inspection and sample alignment

2.6 The first R commands answer structural questions

Structural and alignment checks

```
# Generic feature-by-sample matrix
class(expr)
dim(expr)
expr[1:5, 1:4]
range(expr, na.rm = TRUE)
summary(as.numeric(expr[, 1]))

# Integrity checks
sum(is.na(expr))
anyDuplicated(rownames(expr))
anyDuplicated(colnames(expr))
all(vapply(as.data.frame(expr), is.numeric, logical(1)))

# Metadata alignment
stopifnot(all(colnames(expr) %in% metadata$sample_id))
rownames(metadata) <- metadata$sample_id
metadata <- metadata[colnames(expr), , drop = FALSE]
stopifnot(identical(colnames(expr), rownames(metadata)))
```

Questions to answer before transforming values

Question	Diagnostic clue	Possible action
Are rows features and columns samples?	Dimensions, headers, feature IDs, and sample names.	Transpose only after confirming the file specification.
Are values raw counts?	All non-negative integers; file and methods say counts.	Retain integers for count-based modeling.
Are values already log-transformed?	Continuous values in a narrow range; publication describes log2.	Do not log-transform again.
Are feature IDs unique?	<code>anyDuplicated(rownames(expr)) > 0</code> .	Resolve duplicates with annotation-aware rules; do not silently keep the first.
Are sample IDs unique and complete?	Matrix columns and metadata keys mismatch.	Create a mapping table and stop if mapping is ambiguous.
Are missing values expected?	NA pattern linked to assay or failed sample.	Investigate cause; avoid automatic imputation for count data.

Stop condition: If the matrix orientation, value type, feature identity, or sample mapping is uncertain, pause. Cleaning uncertain data only converts an obvious problem into a hidden one.

Missing values, duplicate identifiers, and impossible entries

2.7 Integrity problems need cause-specific responses

Problem	Why it occurs	Preferred response	Avoid
NA in count matrix	Parsing error, incomplete table, or failed quantification.	Return to source file and metadata; exclude only with documented reason.	Replacing NA with zero without evidence.
Duplicate gene IDs	Version stripping, probe-to-gene mapping, or repeated rows.	Retain original IDs; aggregate only with a justified function and annotation.	make.unique() followed by pretending rows are distinct genes.
Non-numeric columns	Metadata embedded in the matrix or parsing delimiter mismatch.	Separate annotation columns and re-import explicitly.	Coercion that creates silent NAs.
Negative “counts”	Data are transformed, centered, or batch-adjusted.	Reclassify value type; do not use as raw counts.	Rounding to integers.
Duplicated samples	Technical repeat, duplicate file, or repeated column names.	Investigate provenance and biological identity.	Keeping both as independent replicates.
Impossible metadata	Age, sex, stage, or treatment contradicts source records.	Build a discrepancy log and inspect GSM/publication.	Choosing the most convenient label.

R inspection patterns

```
# Missingness summary
missing_by_feature <- rowSums(is.na(expr))
missing_by_sample <- colSums(is.na(expr))
summary(missing_by_feature)
missing_by_sample[missing_by_sample > 0]

# Duplicate identifiers
which(duplicated(rownames(expr)) | duplicated(rownames(expr), fromLast = TRUE))

# Non-finite values
sum(!is.finite(expr))
```

Documentation requirement: Every removed sample or feature class should appear in a cleaning log with the original count, reason, rule, and resulting count.

Filtering low-information features

2.8 Filtering reduces noise and the multiple-testing burden

Features with almost no information across the relevant samples cannot support reliable inference. The threshold should reflect assay type, group size, library depth, and the downstream model. A fixed rule such as “sum ≥ 10 ” is useful for teaching but should not be presented as universally optimal.

Assay / purpose	Reasonable teaching rule	Stronger project-level rule	Caution
Bulk RNA-seq counts	Keep genes with count ≥ 10 in at least the size of the smallest group, or a simple total-count threshold for a toy exercise.	Use <code>edgeR::filterByExpr()</code> with the design or a documented CPM/group rule.	Do not filter using differential p-values.
Microarray exploratory PCA	Remove invariant or lowest-variance probes for visualization.	Use platform-aware detection filtering and retain all appropriate probes for modeled analysis.	Variance filtering changes the displayed structure.
Single-cell RNA-seq	Apply cell and gene QC using counts, detected features, mitochondrial fraction, and sample context.	Use sample-aware thresholds and doublet detection.	Cell-level thresholds are platform- and tissue-specific.
Proteomics / metabolomics	Filter features with excessive missingness or poor QC detection.	Use assay-specific missingness and QC rules.	Missingness may be biologically or technically non-random.

Count-aware filtering

```
# Simple teaching filter
smallest_group <- min(table(metadata$condition))
keep <- rowSums(counts >= 10) >= smallest_group
counts_filtered <- counts[keep, , drop = FALSE]

# Design-aware option in edgeR
library(edgeR)
y <- DGEList(counts = counts, samples = metadata)
keep2 <- filterByExpr(y, group = metadata$condition)
counts_filtered2 <- counts[keep2, , drop = FALSE]

c(before = nrow(counts), after = nrow(counts_filtered2))
```

Report both sides: A reproducible method states the filtering rule and reports the number of features before and after filtering.

Sample-level diagnostics before modeling

2.9 Inspect library size, distributions, and correlations

Diagnostic	What it can reveal	Interpretation caution
Library size / total counts	Failed or shallow libraries; large sequencing-depth differences.	Biological RNA composition can also change total counts.
Number of detected features	Low-complexity or poor-quality samples.	Cell/tissue composition influences detection.
Count distribution / density	Transformation errors or abnormal samples.	Raw counts are expected to be skewed.
Sample correlation heatmap	Replicate similarity, outliers, or batch structure.	High correlation does not guarantee correct labels.
PCA / MDS	Dominant axes of variation and candidate confounders.	PCA is descriptive; it does not prove cause or justify automatic exclusion.
Metadata-colored plots	Association of PCs with condition, batch, sex, tissue, or subject.	Confounded variables may be impossible to separate.

Library-size diagnostic

```
library(ggplot2)
qc <- data.frame(
  sample_id = colnames(counts_filtered),
  library_size = colSums(counts_filtered),
  detected_features = colSums(counts_filtered > 0),
  condition = metadata$condition,
  batch = metadata$batch
)

ggplot(qc, aes(sample_id, library_size, fill = condition)) +
  geom_col() +
  coord_flip() +
  scale_y_log10() +
  theme_bw() +
  labs(x = NULL, y = "Library size (log10)")
```

Outlier rule: A surprising point should trigger investigation of provenance, quality metrics, labels, and sensitivity—not immediate deletion.

PCA, outliers, and batch effects

Batch effects can dominate principal components without changing the biological question

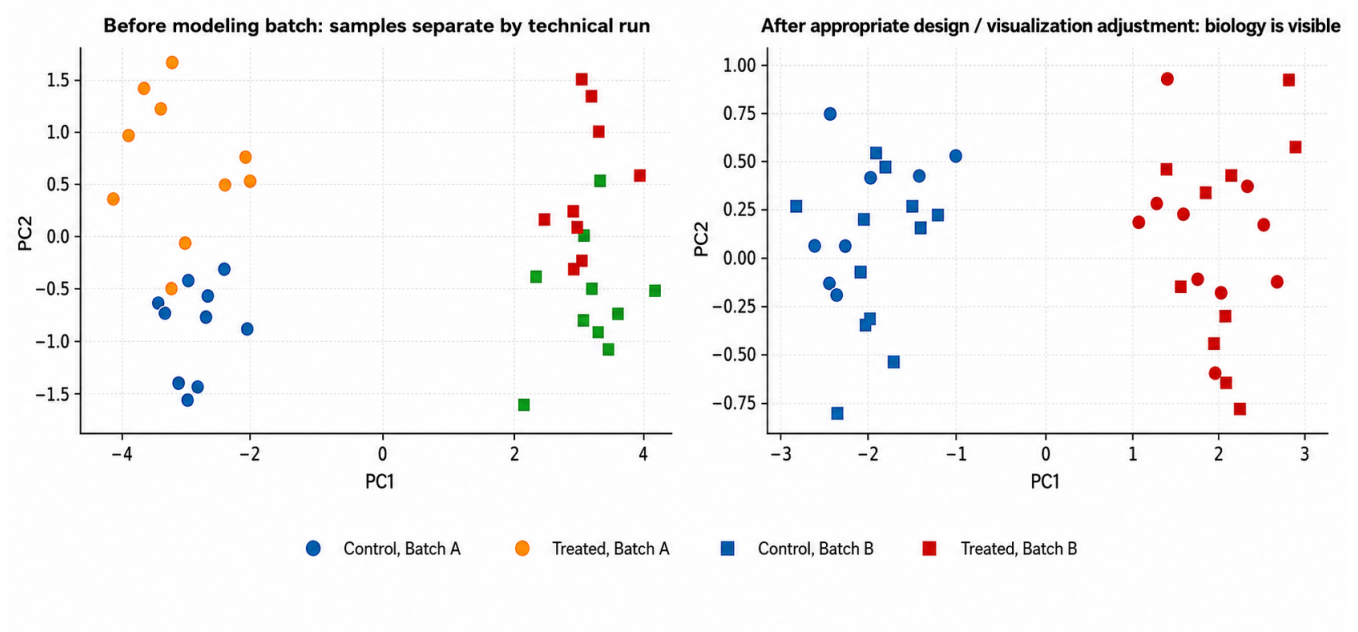


Figure 7. Conceptual PCA example. Technical batch can dominate the first principal component even when the biological question is unchanged.

2.10 Distinguish three different actions

Goal	Recommended approach	Why
Differential testing with known batch	Include batch in the statistical design, e.g., ~ batch + condition.	Models technical variation without replacing observed counts.
PCA or heatmap visualization	Transform counts (VST/log-CPM) and optionally remove batch effect for display, while keeping an unadjusted plot.	Helps visualize biology but must not hide confounding.
Cross-study harmonization	Use carefully justified batch/harmonization methods with sensitivity analyses.	Study effects may include real biological differences.
Batch perfectly confounded with condition	Do not claim statistical correction solves the design.	The data contain no information that separates batch from biology.

Model batch; adjust transformed values only when the purpose is explicit

```
library(DESeq2)

dds <- DESeqDataSetFromMatrix(
  countData = counts_filtered,
  colData = metadata,
  design = ~ batch + condition
)

vsd <- vst(dds, blind = FALSE)
plotPCA(vsd, intgroup = c("condition", "batch"))

# For visualization only, not as replacement counts for DESeq2:
mat_vis <- limma::removeBatchEffect(assay(vsd),
  batch = metadata$batch,
  design = model.matrix(~ condition, metadata))
```

Batch correction is not a substitute for experimental design

2.11 Questions to ask when PCA clusters by batch

1. Is batch recorded correctly, and does it represent run, laboratory, extraction date, plate, or another process?
2. Is batch associated with condition, tissue, subject, or other biological variables?
3. Are there sample swaps, duplicated samples, or inconsistent labels?
4. Can the design estimate both batch and condition, or are they perfectly confounded?
5. Does the biological conclusion persist when batch is included in the model?
6. Are conclusions robust in an independent dataset or balanced subset?

Pattern	What it means	Recommended conclusion
Batch and condition both represented in each group	The model can often estimate both effects.	Include batch in design and report diagnostics.
One condition occurs only in one batch	Perfect or near-perfect confounding.	Do not claim correction isolates biology; find a better dataset or narrow the question.
A single sample is distant in all QC views	Possible quality failure, swap, or true biology.	Investigate raw QC and metadata; perform sensitivity analysis.
Batch pattern remains after modeling	Additional technical variables or nonlinear effects may exist.	Review pipeline and consider surrogate-variable methods with caution.
Correction removes both batch and condition separation	Batch and condition may overlap or adjustment may be too aggressive.	Show before/after plots and revisit design assumptions.

Reporting standard: Show unadjusted QC, state the batch variable and model formula, explain whether batch is confounded, and distinguish model adjustment from visualization-only correction.

Normalization is purpose-specific

Normalization choices depend on data type and analytical purpose

The same matrix should not be transformed identically for differential testing, visualization, and within-sample interpretation.

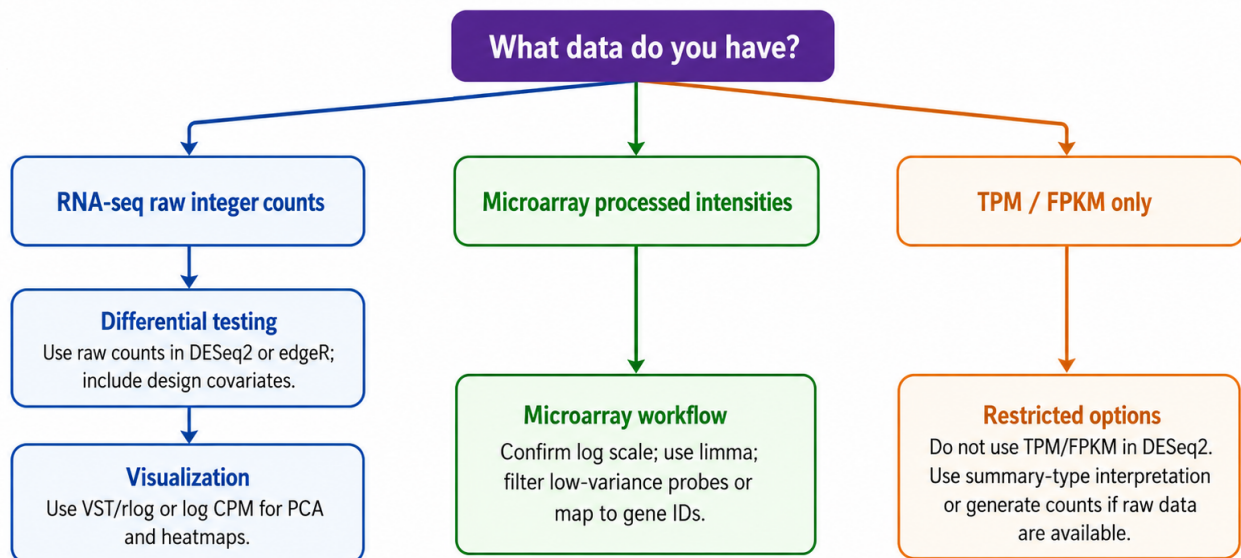


Figure 8. Normalization decisions depend on assay type and analytical purpose.

Input / method	Use	Do not use for
Raw RNA-seq counts + DESeq2 median-of-ratios	Differential expression in the DESeq2 model.	Manual pre-normalization before constructing DESeqDataSet.
Raw RNA-seq counts + edgeR TMM	Differential expression with edgeR/limma-voom workflows.	Combining with DESeq2 size factors in the same model.
VST / rlog / log-CPM	PCA, clustering, sample correlations, and heatmaps.	Supplying transformed values as raw counts to count models.
TPM / FPKM	Within-sample relative expression and selected visualization.	DESeq2 count-based differential expression.
Microarray processed expression	limma after verifying scale and preprocessing.	Count filters or negative-binomial models.
Batch-adjusted transformed matrix	Exploratory visualization or selected integration tasks.	Replacing original counts in an unqualified differential model.

Exam-ready rule: For DESeq2 or edgeR, begin with raw integer counts. For PCA and heatmaps, use a variance-stabilizing or log-count transformation. For microarrays, use limma-oriented continuous-data methods.

AI and agent-assisted data fetching and cleaning

AI and agents can accelerate retrieval and cleaning only when every action is inspectable

The human defines the question, approves the plan, verifies file identity, and interprets quality-control results.

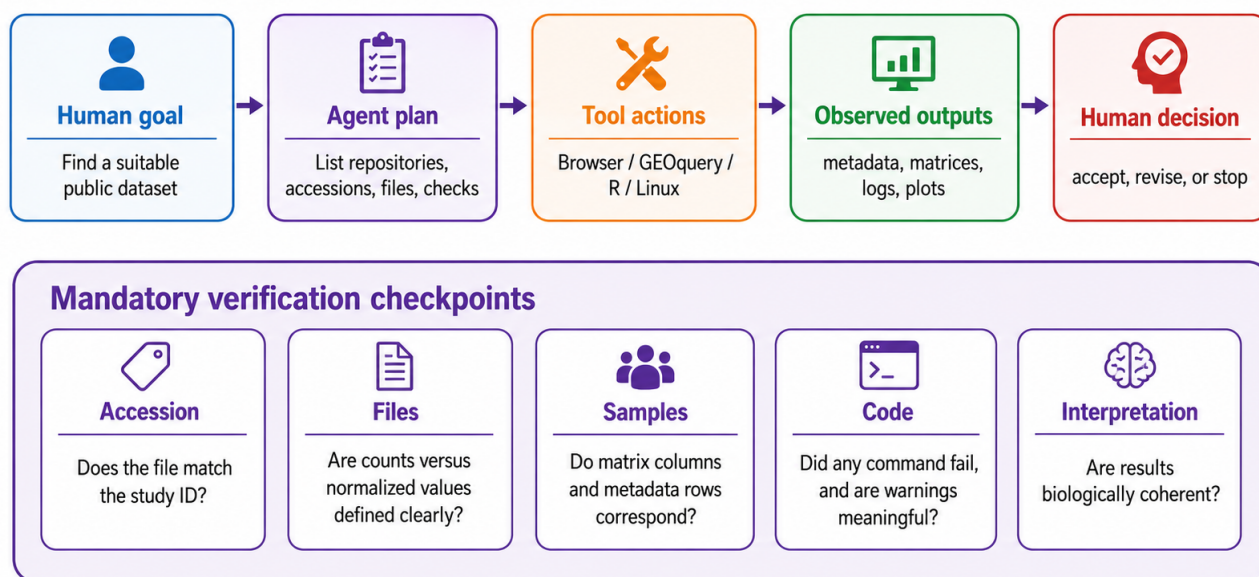


Figure 9. AI and agents can plan and execute technical steps, but the student remains responsible for study selection, permissions, verification, and interpretation.

2.12 Safe division of labor

Role	Appropriate responsibility	Must not be delegated blindly
Student scientist	Define question, approve plan, interpret design, decide exclusions, document limitations.	Final scientific judgment.
LLM / agent planner	Suggest repository, accession checks, file inventory, code structure, and QC checklist.	Inventing file meaning, sample labels, or citations.
CLI / R tools	Execute retrieval, checksums, parsing, transformations, and plots.	Assuming a command succeeded without checking exit status and output.
Reviewer/checker agent	Inspect code, expected object dimensions, warnings, and reproducibility files.	Replacing source verification or biological expertise.

Plan-first prompt: “Do not download or transform anything yet. Restate my biological question, identify the most likely repository and data level, list the accessions and files that must be verified, propose a reproducible folder structure and QC plan, identify possible failure points, and wait for approval.”

Prompt patterns and verification matrix

Task	Useful prompt pattern	Human verification
Repository selection	“Compare GEO, SRA, GDC, GTEx, PRIDE, and MetaboLights for this question. Explain which data object I need.”	Confirm repository scope and access conditions in official documentation.
GEO record interpretation	“Given this copied GSE summary and file list, classify each file as raw, counts, normalized matrix, or derived result. Mark uncertainty.”	Read Series/GSM records and publication methods.
R code drafting	“Write modular R code to retrieve metadata, inspect object class, align IDs, and stop on mismatches. Do not assume column names.”	Run line by line; inspect warnings, dimensions, and files.
Cleaning plan	“Propose checks for missingness, duplicates, low-information features, library size, PCA, and batch. Separate RNA-seq and microarray logic.”	Decide thresholds from design and assay; record before/after counts.
Error diagnosis	“Explain this exact error and propose the smallest test to identify its cause.”	Paste exact code/error; test one hypothesis at a time.
Review	“Audit this script for destructive file operations, hidden paths, unverified assumptions, and missing reproducibility outputs.”	Inspect changes before accepting them.

Verification intensity should match risk

Output	Minimum verification
Explanation of an R function	Read official documentation and run a small local test.
Package or function name	Confirm package documentation and namespace.
Accession, file name, or sample count	Resolve in the official repository record.
Cleaning threshold	Justify from design, assay, and diagnostic plots.
Batch correction decision	Inspect confounding and compare before/after results.
Biological interpretation	Check study design, primary evidence, limitations, and independent support.
Citation / DOI	Open PubMed, DOI, or journal page and verify the claim.

Required AI disclosure: Record the prompt, tool, accepted output, rejected output, verification performed, and remaining uncertainty.

Worked example A | GSE39582 as a processed microarray study

2.13 What the example teaches

GSE39582 is a large colorectal cancer expression cohort measured on the Affymetrix Human Genome U133 Plus 2.0 platform and associated with clinical variables. It is appropriate for learning GEOquery, phenotype-table inspection, probe annotation, log-scale verification, PCA, and limma-oriented analysis. It is not an RNA-seq count dataset [15].

Decision point	Expected observation	Action
Object type	One GEOquery object for GPL570.	Inspect class; use assay()/colData() or legacy accessors.
Rows	Affymetrix probe identifiers, not unique gene symbols.	Use GPL/annotation package; document many-to-one mapping.
Values	Submitter-processed continuous expression values, typically log-scale.	Inspect range and distribution; do not round to counts.
Metadata	Many clinical fields encoded in characteristics columns.	Create a clean data dictionary and explicit sample table.
Filtering	Low-variance probes may be excluded for exploratory PCA.	Report rule; do not claim it is a universal preprocessing requirement.
Differential method	Continuous microarray model.	Use limma, not DESeq2.

Microarray retrieval and exploratory PCA preparation

```
library(GEOquery)
library(SummarizedExperiment)
library(matrixStats)

obj <- getGEO("GSE39582", GSEMatrix = TRUE, getGPL = FALSE)[[1]]
expr <- assay(obj)
pheno <- as.data.frame(colData(obj))

range(expr, na.rm = TRUE)
dim(expr)
head(pheno$title)

# Exploratory variance filter for PCA only
v <- rowVars(expr)
expr_pca <- expr[v >= quantile(v, 0.25, na.rm = TRUE), ]
pca <- prcomp(t(expr_pca), scale. = FALSE)
```

Interpretation boundary: A top-variance PCA is a quality-control view selected from the same matrix. It does not independently validate a clinical subtype or biological mechanism.

Worked example B | GSE87487 as a compact RNA-seq count study

2.14 A count matrix plus paired metadata

GSE87487 contains 20 human liver biopsy RNA-seq samples and a small supplementary count matrix. The overall design includes pre- and post-reperfusion biopsies and IRI status, so the metadata must preserve pairing and multiple factors [6].

RNA-seq retrieval, verification, filtering, and object construction

```
library(GEOquery)
library(DESeq2)

# 1. Inspect the inventory before downloading
supp <- getGEOSuppFiles("GSE87487", fetch_files = FALSE)
supp[, c("fname", "size")]

# 2. Download the confirmed files
getGEOSuppFiles("GSE87487", baseDir = "data_raw")
count_file <- file.path("data_raw", "GSE87487",
                        "GSE87487_counts.20samples.txt.gz")
counts <- read.delim(count_file, row.names = 1, check.names = FALSE)

# 3. Verify count properties
stopifnot(all(as.matrix(counts) >= 0))
stopifnot(all(as.matrix(counts) == round(as.matrix(counts)))))

# 4. Build metadata from the GEO record, then align explicitly
record <- getGEO("GSE87487", GSEMatrix = TRUE, getGPL = FALSE)[[1]]
meta <- as.data.frame(colData(record))
# Hand-curate subject, time_point, and IRI_status after inspecting characteristics.
# Do not infer the design solely from short sample titles.

# 5. Filter and create a design-aware object
smallest_group <- min(table(meta$IRI_status))
keep <- rowSums(counts >= 10) >= smallest_group
counts_f <- counts[keep, ]

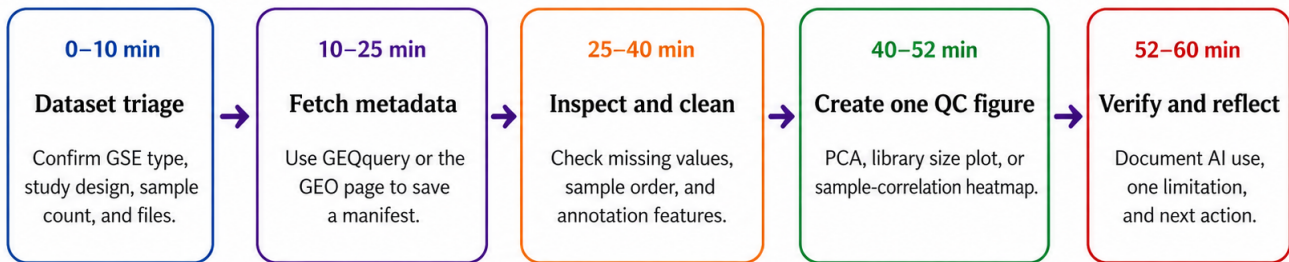
dds <- DESeqDataSetFromMatrix(countData = counts_f,
                              colData = meta,
                              design = ~ subject_id + time_point + IRI_status)
vsd <- vst(dds, blind = FALSE)
```

Do not copy the example formula blindly: The correct design depends on the exact question, pairing, missing combinations, and rank of the model matrix. Students must inspect the metadata and define the contrast before analysis.

One-hour homework | From repository record to QC evidence

The Week 3 homework is designed to fit within one focused hour

Students submit evidence of retrieval, inspection, AI use, and scientific judgment—not a full differential-expression analysis.



Required submission

1) dataset triage card; 2) R script or documented web workflow; 3) one QC figure; 4) cleaned-object summary; 5) 150-word reflection; 6) prompt and verification log.

Figure 10. The homework is scoped to one focused hour and stops before full differential-expression analysis.

Deliverable	Required content	Points
Dataset triage card	Question, GSE, study design, data layer, six criterion scores, proceed/reject decision.	20
Retrieval record	R script or documented web workflow; file inventory; accession; date; file type.	20
Integrity and cleaning summary	Dimensions, sample alignment result, missing/duplicate checks, filtering rule, before/after feature count.	20
One QC figure	PCA, library-size plot, or sample-correlation heatmap with a two-sentence interpretation.	20
AI and verification log	Prompt, accepted/rejected advice, exact source or test used to verify it.	10
Reflection (≤150 words)	Largest limitation, one claim the dataset can support, and one next analysis or validation.	10

Recommended dataset options

- GSE87487 (RNA-seq counts; recommended for students who have GEOquery installed).
- GSE39582 (processed microarray; use the microarray branch and do not apply count-specific rules).
- A different instructor-approved GSE connected to the student's Week 2 research question.

Submission package: week3_homework.R, dataset_triage.pdf or .docx, one PNG figure, cleaning_summary.tsv, and ai_verification_log.txt. The package should run or be inspectable without the instructor guessing what any file means.

Week 4 quiz preparation and self-check

Likely quiz learning targets

Target	A student should be able to...
Repository choice	Select GEO, SRA/ENA, GDC, GTEx, PRIDE, MetaboLights, or MGnify for a stated data need.
Accession prefixes	Distinguish GSE, GSM, GPL, GDS, SRP/BioProject, SRX, and SRR.
Data layers	Distinguish FASTQ, raw counts, TPM/FPKM, processed microarray values, and derived results.
Metadata alignment	Explain why matrix columns and metadata rows must be identical and ordered.
Filtering	Choose a defensible low-information filter and report before/after counts.
Normalization	Match raw counts, VST/log-CPM, TPM/FPKM, and microarray values to appropriate uses.
Batch effects	Recognize confounding and distinguish model adjustment from visualization correction.
AI verification	State what must be verified for accessions, files, code, statistics, biology, and citations.
Reproducibility	Describe raw-data preservation, manifests, scripts, sessionInfo(), renv, and version control.

Suggested reading, references, and preparation for Week 4

Suggested reading sequence

Priority	Reading	Why it matters
Required	NCBI GEO overview, download documentation, and GEO2R caveats [1–3].	Repository navigation, file meaning, and limitations.
Required	GEOquery getting-started vignette and package paper [4,5].	Current R object structure and reproducible retrieval.
Required	Love et al. DESeq2 and Ritchie et al. limma [9,11].	Understand why RNA-seq counts and microarray values use different models.
Recommended	Wilkinson et al. FAIR principles [7].	Connect findability and metadata to reuse.
Recommended	Leek et al. batch effects and Zhang et al. ComBat-seq [12,13].	Recognize confounding and limits of correction.
Recommended	Sandve et al. reproducible computational research [14].	Turn scripts and provenance into scientific evidence.

References

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- [15] Marisa L, et al. Gene expression classification of colon cancer into molecular subtypes. PLoS Medicine. 2013;10:e1001453. doi:10.1371/journal.pmed.1001453.
- [16] Brazma A, et al. Minimum information about a microarray experiment (MIAME). Nature Genetics. 2001;29:365–371. doi:10.1038/ng1201-365.

Pre-reading for Week 4

- Review DNA → RNA → protein and the difference between a reference genome and an annotation.
- Read a short overview of short-read next-generation sequencing and FASTQ quality scores.
- Define SNP, indel, genotype, allele frequency, read depth, and variant call.
- Bring the Week 3 project folder and confirm that R, GEOquery, and the homework script run.